

Multisensory Tiger Moth VR Flight Experience: Work-in-Progress

Danel Ushkempirova*
University of Warwick, UK

Kurt Debattista
University of Warwick, UK

Malcolm Barnes
Hollywood Gaming, UK

Mark Evans
Hollywood Gaming, UK

Alan Chalmers
University of Warwick, UK

ABSTRACT

Current Virtual Reality (VR) systems often provide limited sensory input, reducing the perceived realism of simulations. Research on multisensory VR in flight contexts, particularly including olfactory cues, remains limited. This work-in-progress presents the development of a multisensory Tiger Moth VR flight simulator, detailing its construction, completed real-world data capture, and planned studies to examine how different sensory modalities influence objective and subjective user responses in VR digital tourism experiences.

Index Terms: Multisensory Virtual Reality, Flight Simulator, Digital Tourism, Olfactory Cues

1 INTRODUCTION

Virtual Reality (VR) is rapidly advancing and increasingly applied across diverse domains. Although multisensory input has the potential to enhance VR realism [4], there is still limited understanding of how users perceive additional sensory modalities, such as motion, wind, or smell. A systematic review on Extended Reality (XR) in flight simulators highlights the need for highly realistic, multisensory simulators that closely resemble real-world experiences [1], pointing to a key research gap in immersive VR for flight simulation, especially for open cockpit flights and digital tourism.

This project addresses the gap by experimentally investigating how different sensory cues shape perceived realism within a multisensory Tiger Moth VR flight simulator. The key contribution of this work is an empirical examination of both isolated modality-specific and combined effects of olfactory, motion, and wind cues on presence, subjective stress levels, physiological arousal, motion sickness, and enjoyment in a VR flight experience. Evidence-based recommendations will be provided to support the design of more effective, human-centred multisensory VR systems, particularly in the context of digital tourism. These anticipated findings are expected to have direct implications for high-fidelity pilot training, emergency preparedness, and immersive digital tourism applications.

2 METHODS

2.1 Design

A between-subjects design will be employed with one independent variable (IV), the Level of Realism, consisting of five conditions (Table 1). The dependent variables (DV) are stress levels measured using both objective and subjective measures, as well as presence, motion sickness, user preference and enjoyment ratings. This selection captures both physiological and psychological aspects of the user experience in multisensory VR. Participants will be randomly assigned to one of five conditions and receive a £10 voucher for participation.

*e-mail: Danel.Ushkempirova@warwick.ac.uk



Figure 1: Real Tiger Moth flight and data capture setup: (a) Tiger Moth biplane used for visual recordings and the preliminary study; (b) real aircraft instrument panel; (c) Insta360 Pro II camera mounted on a mannequin pilot.

Table 1: Study experimental conditions

Condition	Description
VR	Traditional VR (audio-visual)
VRS	Traditional VR with olfactory cues added (audio-visual-olfactory)
VRM	Traditional VR with motion cues added (audio-visual-motion)
VRW	Traditional VR with tactile wind cues added (audio-visual-tactile)
VRSMW	Full multisensory VR (audio-visual-olfactory-motion-tactile)

The research question is: What are the isolated and combined effects of different sensory modalities on objective and subjective measures of realism in a multisensory VR flight simulation? The proposed hypotheses are:

H1: The full multisensory simulation will elicit significantly higher stress levels (objective and subjective), presence, and enjoyment compared to lesser sensory simulations.

H2: Different sensory modalities will have distinct effects on physiological and psychological experience.

H3: Adding olfactory, motion, or wind cues individually to the VR simulation will increase perceived realism compared to audio-visual only.

2.2 Real Flights for Data Capture

The VR simulation was informed by a real-world data capture session in which five volunteers flew a real Tiger Moth. The session was conducted at Sywell Aerodrome, UK, in collaboration with Thomas Castle Aviation Heritage Ltd on 29 June 2022. The aircraft used for visual and sensory recordings is shown in Figure 1a.

Physiological and psychometric data were collected to characterise the real flight experience and inform VR simulator design. Physiological data were recorded using a VU-AMS device. Although highly accurate, participant movement occasionally caused

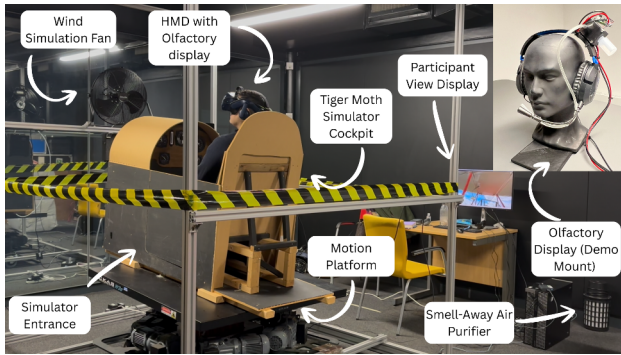


Figure 2: Finished multisensory Tiger Moth VR simulator with key components annotated.

electrode detachment and data loss. To address this limitation, validated commercial wearables will be used in the main study.

2.2.1 Data Capture

The 360° video was captured using an Insta360 Pro II 360 VR camera mounted on a custom-built human mannequin dressed in pilot-style clothing, representing the participants' virtual body in the simulation (Figure 1c). Telemetry data, including accelerometer readings, were recorded simultaneously using Android HTC One M8 accelerometers mounted on the mannequin to capture motion parameters during flight. Volunteers' free descriptions of the experience were used to select olfactory stimuli for validation in the pilot study.

2.3 Tiger Moth Simulator

Visuals. The footage captured during the real flight was rendered at 4K, 2D, 360° for the VR study, with a total duration of 6 minutes, and displayed using an HTC Vive Pro head-mounted display.

Apparatus. The aircraft used was G-ANTE, a DH-82A Tiger Moth built for the Royal Air Force in 1941 (serial T6562; Figure 1a). The experimental apparatus comprises a CKAS W3 six-degrees-of-freedom (6DOF) motion platform with a replica cockpit mounted on top, including a fixed aircraft-style seat with seat-belt (Figure 2). The cockpit features an instrument panel replicating standard analogue flight instruments, including an airspeed indicator, altimeter, turn-and-slip indicator, tachometer, oil pressure gauge, and variometer, based on the real Tiger Moth (Figure 1b). The apparatus and additional stimuli are supported by an external aluminium frame measuring 2.4 m × 2.0 m (base) × 2.4 m (height).

Olfactory Display. A wearable olfactory display has been developed by the research team, mounted on the Head-Mounted Display, and synchronised with the other stimuli. The system uses an ultrasonic transducer to atomise oil-based scents delivered to the transducer via capillary action, which are then directed towards the participant's nose by a small fan. AromaPrime scents (cut grass and machine oil), previously validated by the research team, will be used. The olfactory display will be evaluated during a pilot study, assessing scent congruency, intensity, delivery latency, and residual duration. To mitigate carryover effects, a SmellAway air purifier and room ventilation will be employed, with procedures refined based on pilot study outcomes.

Wind Stimuli. Wind stimuli are delivered using a Devola high-velocity floor fan (DVFF20) synchronised with the visual, motion and smell components of the simulation.

2.4 Perception Measures

Both subjective and objective measures will be collected to capture experiential, physiological, and comfort-related responses, en-

abling assessment of how sensory modalities shape user perception of multisensory VR flight. Physiological stress will be assessed via heart rate and electrodermal activity using a Fitbit Sense 2 smart-watch, previously validated against research-grade equipment and used in VR research [2, 7]. The device is comfortable to wear and suitable for repeated use. Psychometric assessments will include the Simulator Sickness Questionnaire for motion sickness [3], the Igroup Presence Questionnaire for presence [5], the State-Trait Anxiety Inventory for stress [6], User Preference Ratings (1–100), enjoyment ratings, and the Immersive Tendencies Questionnaire to measure individual propensity for immersion [8].

2.5 Pilot Study

A pilot study will be conducted to verify that the stimuli and procedures reliably produce the intended perceptual effects. It will aim to assess subjective perceivability, intensity, and congruency, examine potential smell carryover effects, and ensure the smell elimination procedure and overall experience are appropriate and comfortable. Perceived timing alignment across visual, motion, wind, and olfactory modalities will be assessed to verify synchronisation accuracy. The pilot study will also explore individual differences in scent familiarity and congruency.

3 CONCLUSIONS AND FUTURE RESEARCH

The multisensory Tiger Moth VR simulator is complete, and the study has full ethical approval (BSREC 65/24–25). Following the pilot study, 120 participants will be recruited to balance statistical power and feasibility. The findings will clarify the role of different sensory cues in VR flight experiences and inform the design of more realistic digital tourism simulations, with future work enabling real–virtual comparisons via additional Tiger Moth flights.

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REFERENCES

- [1] J. Cross, C. Boag-Hodgson, T. Ryley, T. J. Mavin, and L. E. Potter. Using extended reality in flight simulators: a literature review. *IEEE transactions on visualization and computer graphics*, 29(9):3961–3975, 2022.
- [2] S. Irshad, A. Perkis, and W. Azam. Wayfinding in virtual reality serious game: An exploratory study in the context of user perceived experiences. *Applied Sciences*, 11(17):7822, 2021.
- [3] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The International Journal of Aviation Psychology*, 3(3):203–220, 1993. doi: 10.1207/s15327108ijap0303_3
- [4] M. Melo, G. Gonçalves, P. Monteiro, H. Coelho, J. Vasconcelos-Raposo, and M. Bessa. Do multisensory stimuli benefit the virtual reality experience? a systematic review. *IEEE transactions on visualization and computer graphics*, 28(2):1428–1442, 2020. doi: 10.1109/TVCG.2020.3010088
- [5] T. Schubert, F. Friedmann, and H. Regenbrecht. The experience of presence: Factor analytic insights. *Presence: Teleoperators & Virtual Environments*, 10(3):266–281, 2001. doi: 10.1162/105474601300343603
- [6] C. D. Spielberger, F. Gonzalez-Reigosa, A. Martinez-Urrutia, L. F. Natalicio, and D. S. Natalicio. The state-trait anxiety inventory. *Revista Interamericana de Psicologia/Interamerican journal of psychology*, 5(3 & 4), 1971.
- [7] N. Van Oost, F. Masci, A.-M. Schyvens, B. Peters, H. Dirix, V. Ross, G. Wets, J. Verbraecken, A. Neven, and J.-M. Aerts. A new perspective on validation of wearables for stress monitoring of occupational drivers. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1–4. IEEE, 2024.
- [8] B. G. Witmer and M. J. Singer. Measuring presence in virtual environments: A presence questionnaire. *Presence*, 7(3):225–240, 1998. doi: 10.1162/105474698565686